

Global Conflicts: Impact on Supply Chains

AIS dark spots, chokepoint exposure and the maritime cost of conflict

Competition	College of Defence Management - DATATHON 2026 (under HQ IDS)
Theme	6.2 Global Conflicts - Impact on Supply Chains
Primary datasets	Middle East weekly conflict aggregate (ACLED-style); Sentinel-1 SAR vessel detections with AIS correlation
Observation windows	Conflict: Dec 2014 to 27 Jun 2026 (149,825 rows). Detections: 1-14 Mar 2026 (104,840 rows, 2,830 radar scenes)
Tooling	Python 3.11 (pandas, numpy, scipy, statsmodels, scikit-learn, matplotlib), Power BI star-schema export with DAX mea
Deliverables	This report; 20 figures; 30+ audit tables; Power BI model (outputs/powerbi) with measures.dax and build guide
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The central claim

Conflict does not degrade shipping in one way. It degrades it in two, and they need opposite responses. Where a belligerent can threaten hulls, ships **leave** - the Red Sea corridor is running 81% below the large-vessel throughput of uncontested corridors. Where a belligerent instead needs its own traffic to keep moving unobserved, ships **stay and go dark** - 84% of SOLAS-class vessels in the Strait of Hormuz carried no usable AIS identity, against a 10% open-ocean baseline. A staff that watches only AIS volume sees the first and is blind to the second; a staff that watches only dark rates sees the second and mistakes the first for calm.

1. Executive summary

This analysis joins two supplied datasets that have never been read against each other: 149,825 weekly conflict records for the Middle East (614,134 events and 552,727 fatalities, Dec 2014 to 27 Jun 2026), and 104,840 Sentinel-1 radar vessel detections taken worldwide over fourteen days in March 2026, each carrying the result of an attempt to correlate it with the AIS picture. Radar sees hulls; AIS sees declarations. Where the two disagree, something is being concealed, lost, or simply never transmitted - and the analysis is the business of telling those three apart.

- **The identification crisis is real and it is concentrated.** Across all traffic 39% of detections are unattributed, but for SOLAS-class hulls (≥ 100 m, AIS carriage mandatory) the figure is 15%. That average conceals the finding: the Strait of Hormuz runs at 84% (1,304 large vessels), the Black Sea at 48%, the Gulf of Oman at 46% and the Persian Gulf at 42%, against an open-ocean baseline of 10%.
- **Dark is not one phenomenon.** Of 1,038 half-degree cells with enough traffic to test, 142 are significantly darker than their own traffic mix predicts. Separating them by whether the AIS picture reaches that water at all yields 93 *behavioural* dark spots (selective switch-off by large hulls while others match normally), 106 non-carriage areas (small craft that were never obliged to transmit) and 9 true dead zones (reception or feed gaps, where nothing at all matches). Only the first is an adversary decision.
- **Avoidance is the other failure mode.** Bab-el-Mandeb yielded 56 large-vessel detections in fourteen days against 795 in the Dover Strait; normalised for satellite revisit it runs 81% below the uncontested-corridor benchmark, the Gulf of Aden 81% below. The Cape of Good Hope route carries 21.5 large vessels per imaged scene against 19.4 for the entire Suez/Red Sea route - a detour of roughly 3,500 nautical miles now moves as much large-vessel traffic as the canal corridor it replaced.
- **Conflict and darkness are statistically linked at the level of the individual ship.** In a logistic model over 39,876 SOLAS-class detections, controlling for hull length and fishing behaviour, a vessel within 150 km of recent conflict has 2.39 times the odds of being dark (95% CI 1.24-4.60, $p=0.009$). The corridor-level cross-section is consistent in sign but not significant (Spearman $\rho=0.28$, $p=0.18$) - with 25 corridors it is underpowered, and it is reported rather than hidden.
- **Dark spots are predictable enough to be planned against.** A gradient-boosted classifier over cell-level traffic, geography and conflict features, validated on *whole 5-degree spatial blocks it never saw in training*, reaches ROC-AUC 0.80 and PR-AUC 0.35 against a base rate of 0.09. Distance to the nearest recent conflict is the second strongest predictor.
- **There is no warning time in the sea/land relationship.** Cross-correlation of littoral violence against ACLED's own maritime-domain events peaks at lag 0 weeks ($r=0.63$) and is nearly flat either side. Attacks at sea move *with* the land campaign, not after it, so maritime posture cannot be sequenced behind a land indicator; it has to move at the same time.
- **Ranked exposure.** The composite Maritime Supply-Chain Risk Index places 6 corridors in the RED tier, led by Gulf of Aden (MSRI 0.72). Weighted by Indian trade dependency, the top exposure is Strait of Hormuz (0.62) - the corridor through which the bulk of India's crude and LPG imports must pass, and the one where the recognised maritime picture has most thoroughly collapsed.

Recommendations in one line each

- Treat a SOLAS-class dark rate above 35% in a corridor as a **reportable maritime indicator** in its own right, logged weekly like any other warning indicator - the measurement needs one radar product and one AIS feed.
- Fuse SAR with AIS rather than choosing between them: **every conclusion here is invisible to either source alone.**

- Separate dead zones from dark behaviour **before** tasking against them; nine cells here would have sent an ISR asset to chase a receiver gap.
- For Indian shipping, plan Hormuz for **identification failure** (independent tracking, escorted transit windows, pre-sailing reporting) and the Red Sea for **absence** (routing, insurance and schedule resilience). They are not the same problem and the same measure does not fix both.
- Stand up the dark-spot classifier as a weekly product: it needs only the previous fortnight's detections and the conflict feed that already exists.

2. Problem definition

The theme asks participants to derive insights into the impact of conflict on global shipping, to identify AIS dead zones, to predict AIS dark spots, and to correlate those with the identification crisis produced by localised conflicts of the Black Sea or Persian Gulf type. Turned into questions a pipeline can answer:

#	Question	Instrument used	Where answered
Q1	How has the conflict environment itself evolved over time?	Weekly aggregation event-type video	Section 4.1, Fig 14
Q2	Where does the AIS picture fail, once vessel size is standardised?	Wilson intervals, BIS	Section 5.2, Fig 5-6, 15
Q3	Which failures are dead zones and which are dark spots?	Cell-type typology using matched	Section 5.3, Fig 16
Q4	Where has conflict removed traffic instead of revealing it?	Revised throughput against unconflicted	Section 6, Figs 7-8, 19
Q5	Is darkness actually related to conflict, or only to detection?	Logistic regression, corridor	Section 6, Fig 18, 20
Q6	Can dark spots be predicted rather than only observed?	Grid-based with spatially blocked cross	Section 6, Fig 12
Q7	What should a commander do first?	Composite MSRI with declared weights; India	Section 6, Fig 13-14

3. Data and preprocessing

3.1 What the two datasets actually contain

Dataset	Grain	Rows	Span	Key fields
Middle East conflict aggregate	week x country x admin1 x event	149,825	Dec 2014 - 27 Jun 2026	events, fatalities, population exposure, admin1
Sentinel-1 SAR vessel detections	one radar contact	104,840 (of 107,257)	14 Mar 2026	position, presence score, length, MMSI, match

The conflict extract is Middle East only, so the coupling analysis is honest only for corridors inside or downstream of that theatre. Corridors elsewhere enter the analysis as controls, which is exactly what an uncontested baseline requires - and is a stronger design than a global conflict feed would have given, because the control group is genuinely unexposed.

3.2 The cleaning decision that changes the answer

The detection file contains two partially contradictory accounts of whether a ship was identified. **matched_category** is the provider's adjudication: 'unmatched' means the correlation was rejected. **mmsi** and **matching_score** describe the candidate association. They disagree in both directions: 13,057 detections carry a candidate MMSI and are still labelled unmatched (a candidate was considered and rejected), while roughly a third of accepted matches sit below a correlation score of 1.0.

Taking 'has an MMSI' as the definition of identified - the obvious first move - puts the global dark rate at 26.6%; taking every weak score as dark puts it at 58.4%; the provider's own label puts it at 39.1%. The pipeline adopts the provider label as the headline, records the other two as explicit alternatives, and re-runs every corridor result under all three (*outputs/tables/darkspot_sensitivity.csv*). The corridor ranking is stable across all three definitions; that stability, not any single number, is what makes the ranking trustworthy.

3.3 Quality gates and derivations

Step	Rule	Effect	Why
Detection confidence	drop presence_score < 0.90	-2,412 rows	Below this the contact is as likely to be clutter, wind streaks or a
Minimum size	drop length < 10 m	-8 rows	Below reliable Sentinel-1 IW detection size
Time parsing	strip ' UTC', parse to tz-aware	0 dropped	Enables joins to the conflict calendar
Dark definition	matched_category == 'unmatched'	39.1% dark	The provider's adjudication, not a re-derivation from a candidate
Size class	6 length bands, SOLAS cut at 100-m		A dark rate is only comparable within a size class
Corridor tagging	28 priority-ordered lane boxes	33% of detections in a narrow corridor	Narrow corridors claim a point before the basin around them
Conflict coordinates	range check on centroids	0 dropped	Guards a silent projection error
Conflict calendar	derive year, month, ISO week, lethality		Common grain for joins and for the Power BI date table

4. Analytical methods and why each was chosen

Problem	Method	Why not the obvious alternative
Dark rates are not comparable between countries	Relative standardisation: expected dark count per vessel from global baseline	A new dark rate table is the baseline for all countries
Small denominators manufacture dark spots	Wilson score intervals on every proportion	The normal approximation is badly wrong at $n < 50$ and would create dark spots
Thousands of cells tested at once	One-sided exact binomial test with Benjamini-Hochberg FDR control at 1/338.05	11s would return roughly 50 false dark spots
Dark spots are areas, not points	DBSCAN on the haversine metric, 60 km neighbourhood, distance to 50 AIS clusters	Light means you don't pick up 50 AIS clusters of a fixed count on a problem
Association is not causation, and detection level fogged by vessel size	Detention level fogged by vessel size with cluster analysis	Correlation does not imply causation
Gridded geodata leaks across borders	Spatially blocked cross-validation on whole world	Salegorn's old leaves neighbouring cells of the same dark spot on both sides
A forecast needs an honest error	Expanding-window backtest of four models; A single day - 5 day interval	Often, a single day interval is not enough to test a model's own performance
Findings must become one decision	Composite MSRI, five min-max scaled components, weights determined by the knowledge of the	Findings, weights determined by the knowledge of the

5. Findings

5.1 The conflict environment, and its move onto the water

The theatre carries 614,134 events and 552,727 fatalities over Dec 2014-27 Jun 2026. Two structural features matter for shipping. First, the mix has shifted towards explosions and remote violence - the category that includes air and drone strikes and anti-ship missile employment - which is the class of violence that reaches ships without any ground advance. Second, and more directly, ACLED now places events on the water itself: 1,257 events fall in the North Indian Ocean, the Wider Black Sea Region or the Eastern Mediterranean, with the series taking off from late 2023. The single heaviest week in the entire record is 03 Jan 2026 (27,696 fatalities, concentrated in Iran), which matters here because internal instability in a littoral state is precisely the condition under which that state's own shipping stops declaring itself.



Figure 1. Weekly events and fatalities, 2015-2026. Events and fatalities are plotted in separate panels rather than on twin axes: two scales on one frame is the most common way a chart misleads.

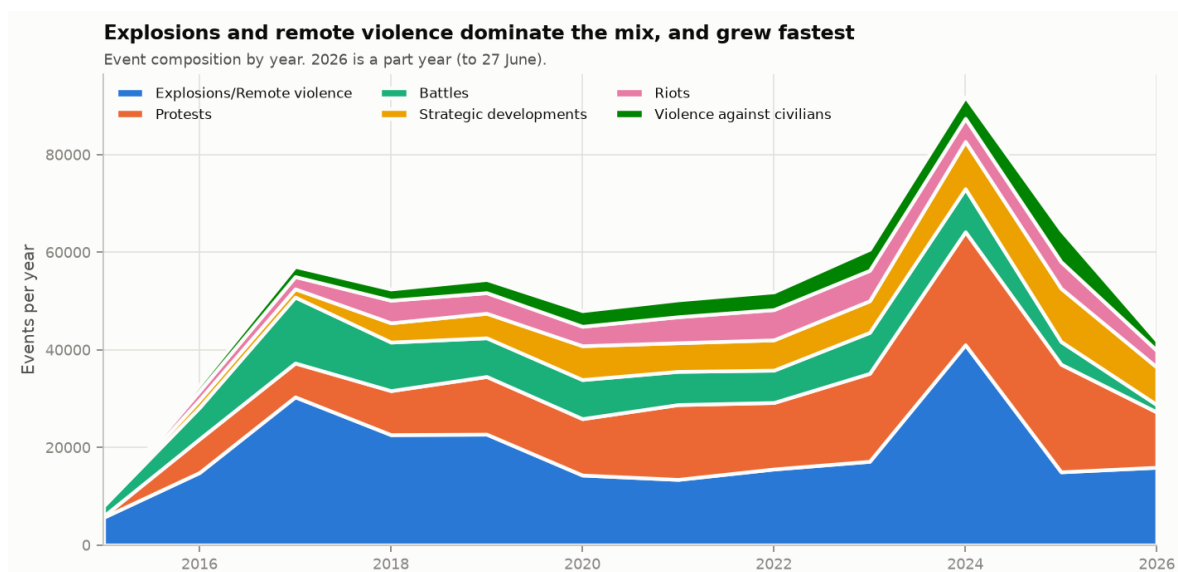


Figure 2. Event composition by year. Explosions and remote violence dominate and grew fastest - the class of violence that reaches shipping without a ground advance.

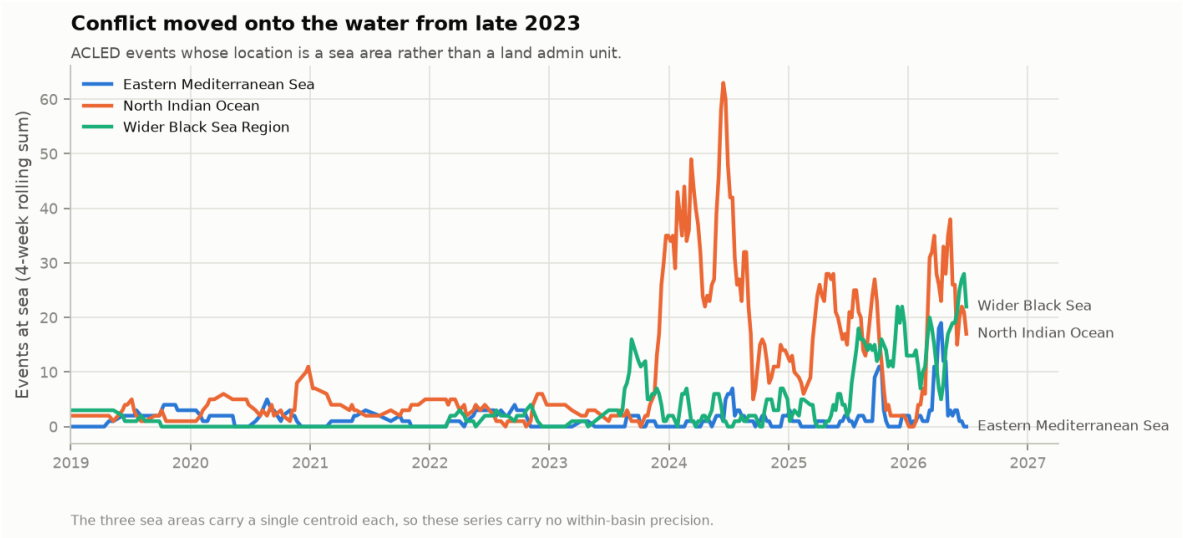


Figure 4. Events ACLED places at sea rather than on land, 4-week rolling sum. Each sea area carries a single centroid, so these series carry no within-basin precision - they establish timing, not position.

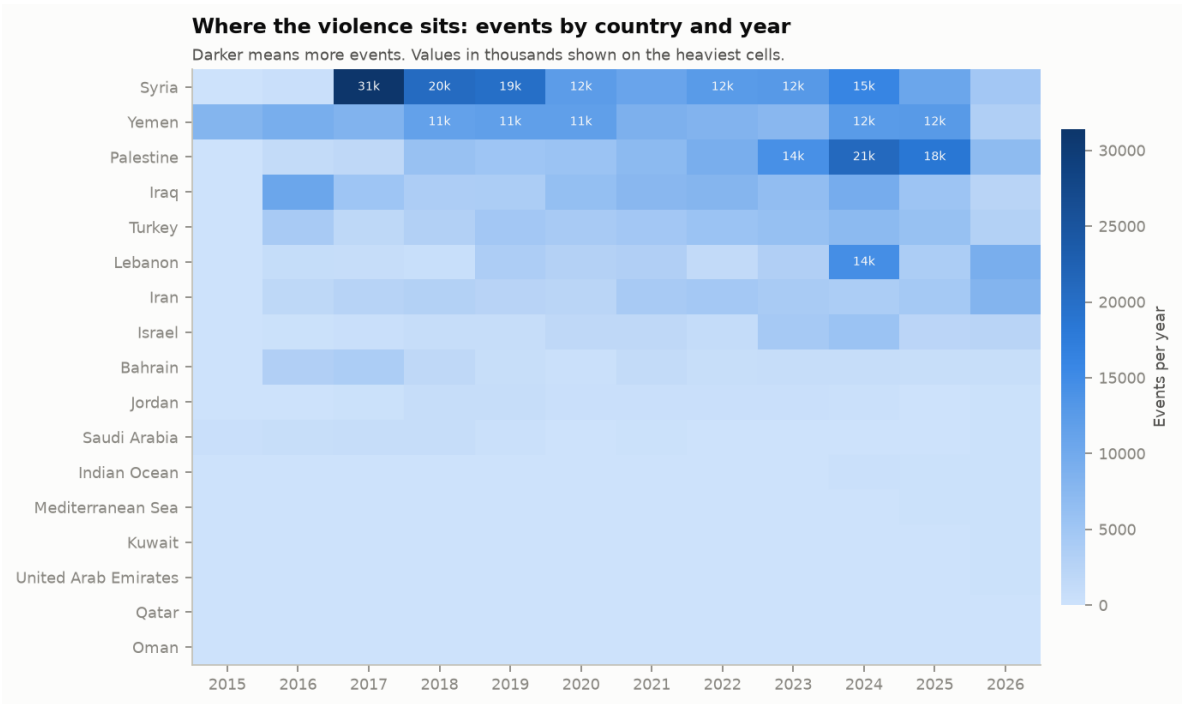


Figure 3. Events by country and year. The 2026 rows cover a part year (to 27 June).

5.2 Why a raw dark rate is a trap

Across the whole detection set 39.1% of contacts are unattributed, and that number is nearly useless. Dark rate falls monotonically with hull size - 75% for craft under 25 m, 13-17% for vessels over 100 m - because below the SOLAS carriage threshold a vessel is not required to transmit at all. Any corridor comparison that ignores this is a map of where the fishing fleets are. Every corridor result in this report is therefore computed on SOLAS-class hulls (39,876 detections, 15.2% dark overall), and cross-checked with size-standardised rates.

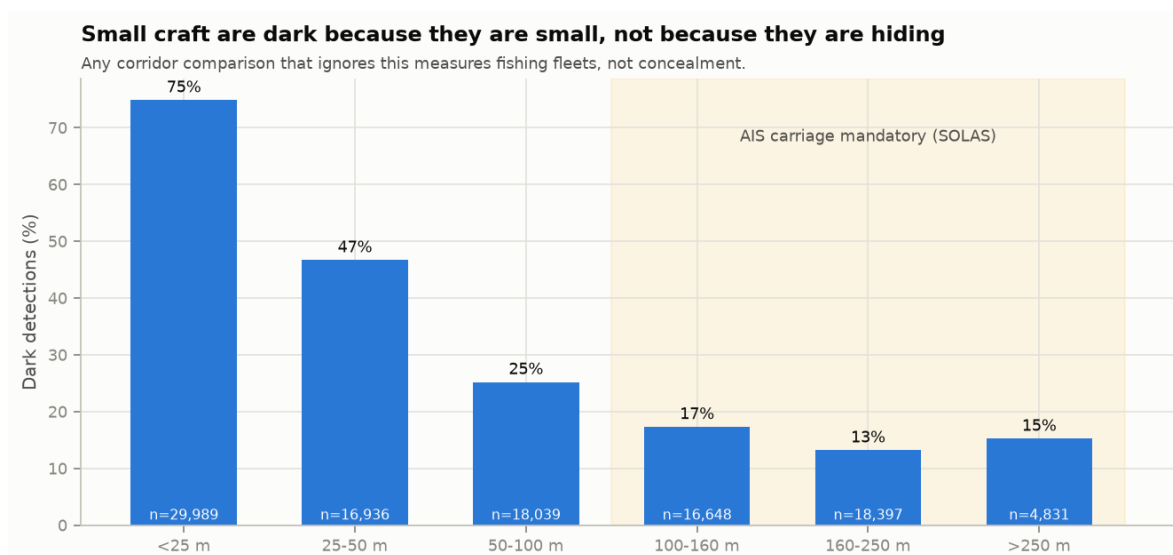


Figure 5. Dark rate by vessel length class. The shaded band marks hulls for which AIS carriage is mandatory; only there does a dark detection mean something is wrong.

5.3 Where large ships stop being identifiable

Against an open-ocean baseline of 10%, four corridors stand far outside the distribution: the Strait of Hormuz at 84% (1,304 SOLAS detections), the Black Sea at 48% (1,465), the Gulf of Oman at 46% and the Persian Gulf at 42%. These are not marginal excursions: the Wilson intervals do not come close to the baseline. The corridors are also precisely the two theatres named in the theme statement - the Persian Gulf and the Black Sea - which is a useful external check that the instrument is measuring what it claims to.

The contrast inside a single theatre is as informative as the ranking. The Turkish Straits, under continuous VTS control, sit at 2%; the Black Sea beyond them at 48%. The Suez Canal, transited under authority supervision, sits at 3% while the Gulf of Oman approach sits at 46%. Darkness is not a property of a region. It is a property of whether a ship expects to be held to account in that water.

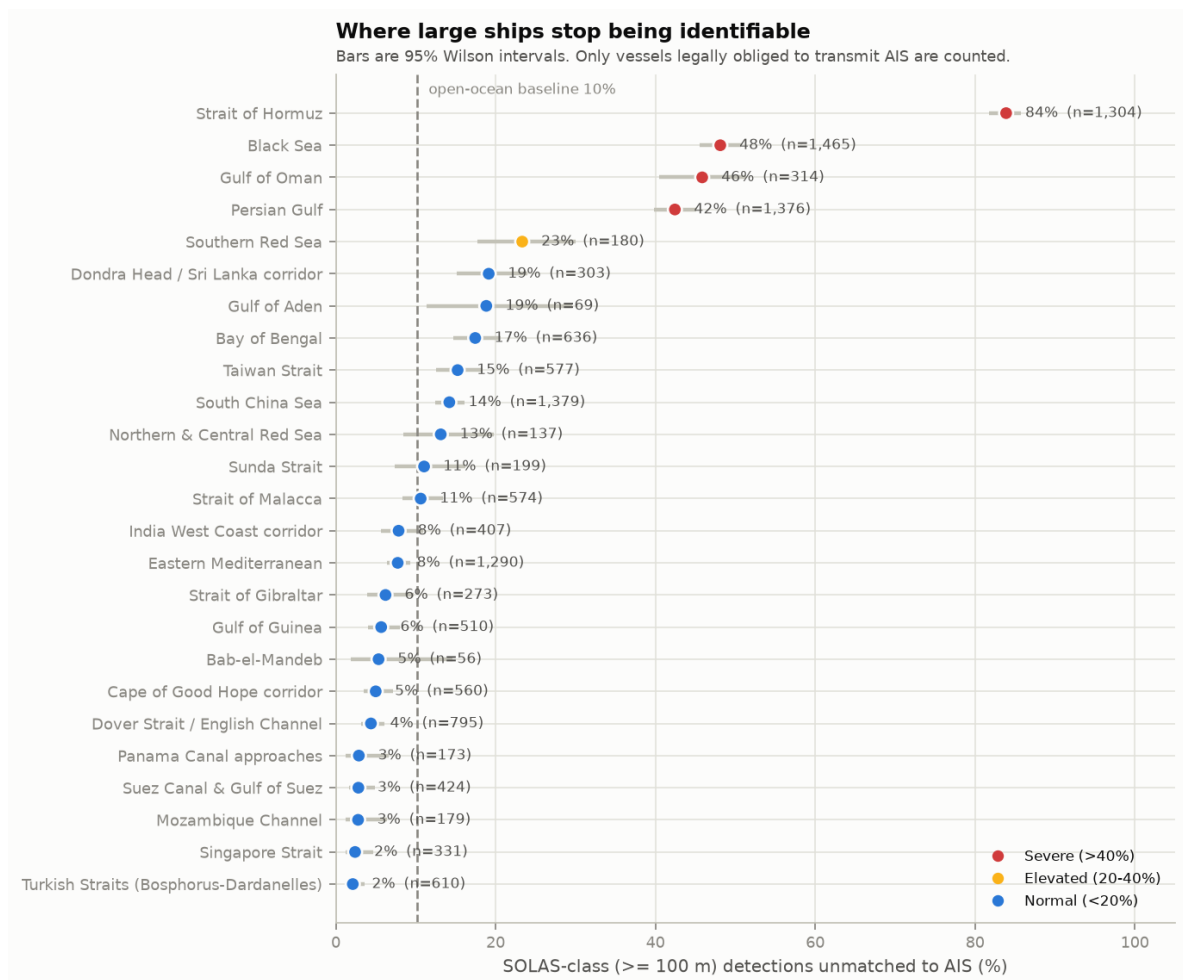


Figure 6. SOLAS-class dark rate by corridor with 95% Wilson intervals. Colour encodes severity band and is always accompanied by the printed value and sample size.

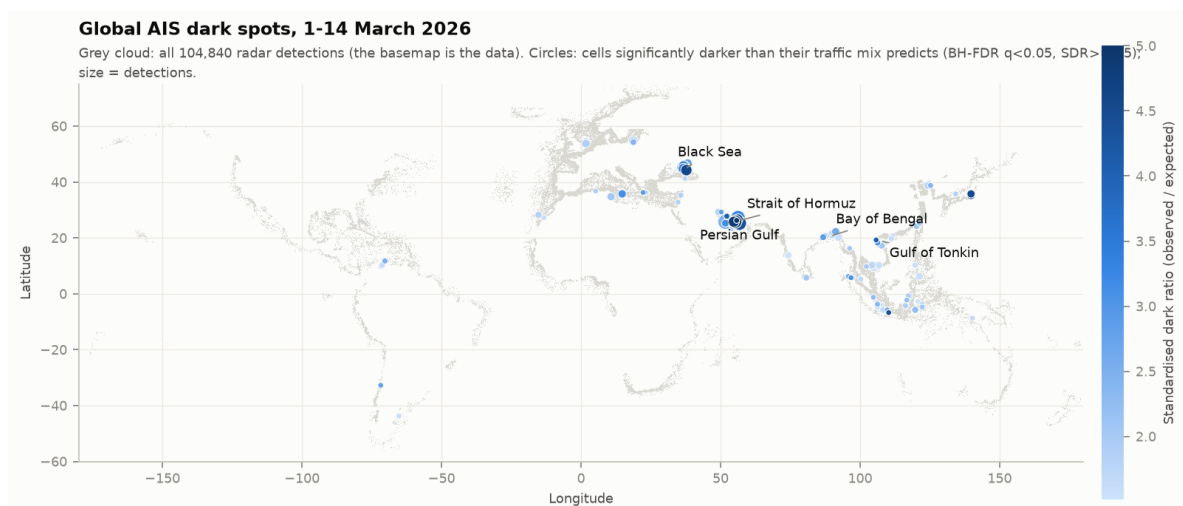


Figure 15. Cells significantly darker than their own traffic mix predicts (BH-FDR $q < 0.05$, $SDR \geq 1.5$). The grey cloud is the full detection set - with 104,840 contacts the data draws its own coastlines, so no basemap dependency is needed.

5.4 Dead zone, non-carriage, or decision? A typology

The theme asks for dead zones and for dark spots as though they were the same object. They are not, and confusing them wastes collection. A **dead zone** is a hole in the AIS picture - no receiver in range, no correlated satellite pass, a feed outage - and reception failure is indiscriminate, so nothing in the cell matches, whatever its size. A **behavioural dark spot** is a decision: some ships in the same water match perfectly well, which proves the picture reaches it, while the large hulls do not.

Applying that discriminator to 1,038 tested cells yields 93 behavioural dark spots, 106 non-carriage areas (small-craft populations that were never obliged to transmit - the Bangladesh and Myanmar coasts, the Gulf of Thailand) and 9 genuine dead zones. Two of the dead zones sit over Tokyo Bay, one of the most densely instrumented waters on earth, where every detection in the cell is unmatched: that pattern is a feed gap, not a fleet acting in unison. Tasking an ISR asset against it would have been a wasted sortie, and telling the two apart costs nothing but the question.

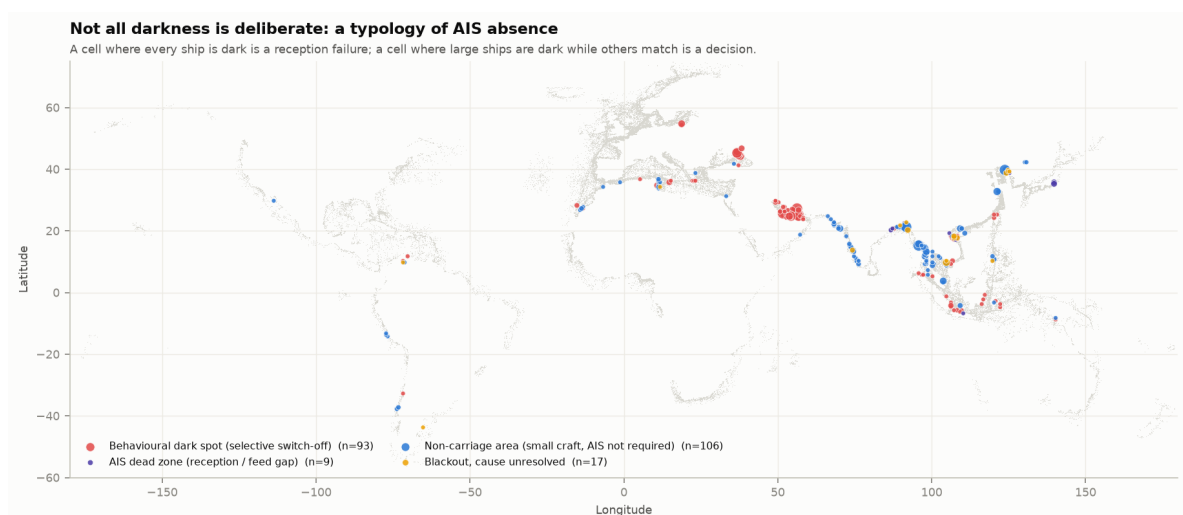


Figure 16. The typology mapped. Categories use an all-pairs colour-validated set and each carries a text label; the classification rule itself is in `analysis/darkspots.py`.

5.5 The two case studies the theme names

Strait of Hormuz. 84% of SOLAS-class detections carry no usable AIS identity, and the pattern is not a thin scatter: DBSCAN recovers a single contiguous dark operating area of 1,401 large dark contacts centred on the strait, with a local dark rate of 82%. Matched and dark vessels occupy the *same* water on the *same* days -

the reception argument fails here in the most direct way available. This is the identification crisis the theme postulates, and it sits on the corridor through which the bulk of India's crude imports pass.

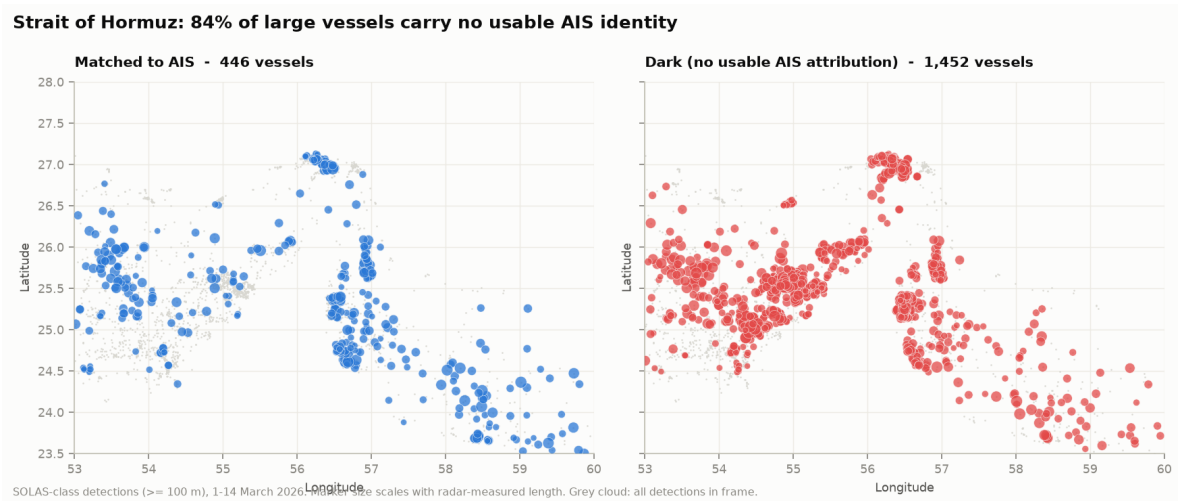


Figure 17. Hormuz, matched against dark SOLAS-class traffic, same water and same fortnight. The dark population is not displaced from the matched one; it is interleaved with it.

Black Sea. 48% dark among large hulls, concentrated on the eastern approaches and around the Kerch Strait, while the Bosphorus - a few hundred kilometres away under Turkish VTS - runs at 2%. The same sea, the same fortnight, the same satellite, a twenty-fold difference in identification. Localised conflict produces a localised identification crisis with a sharp geographic edge, and that edge is drawn by control, not by physics.

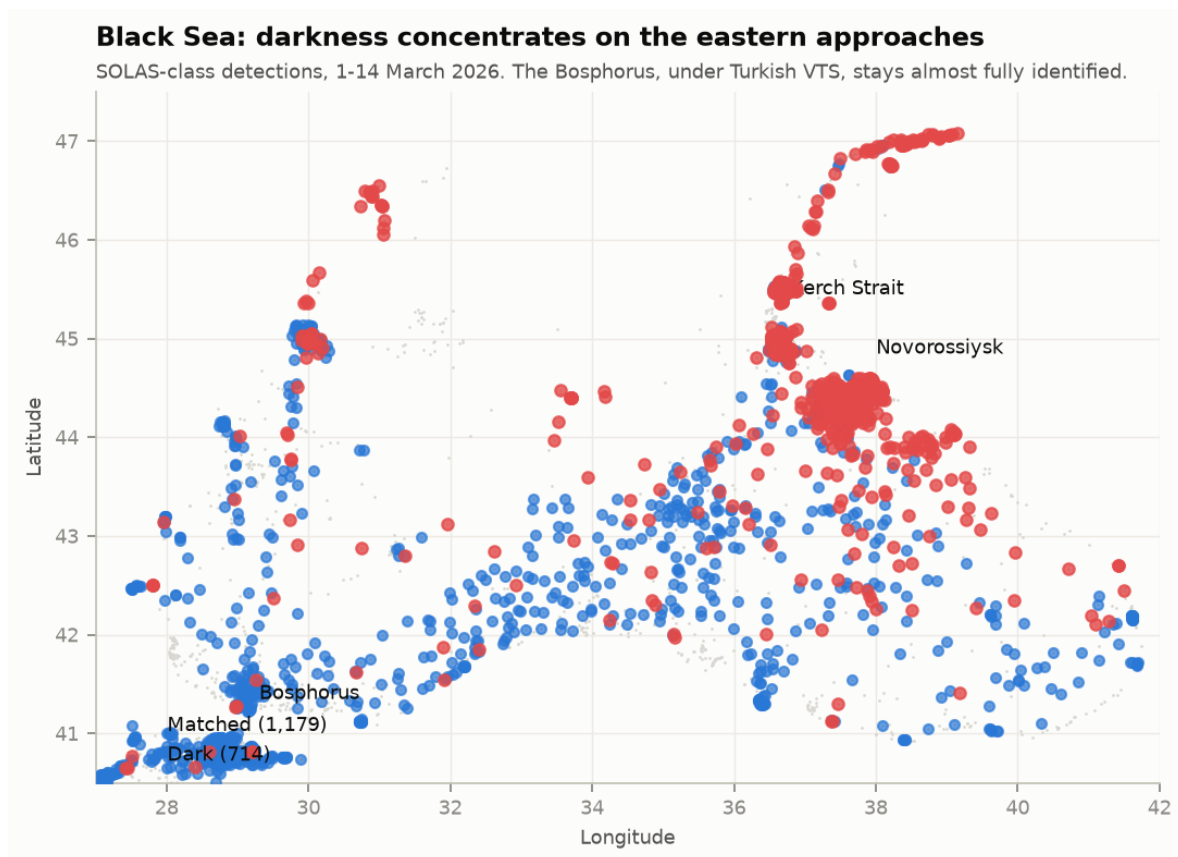


Figure 18. Black Sea SOLAS-class traffic. Darkness concentrates east of the Crimean peninsula; the regulated strait at the exit stays almost fully identified.

5.6 The other failure mode: traffic that leaves

Bab-el-Mandeb produced 56 SOLAS-class detections in fourteen days. The Dover Strait produced 795. Normalised for how often Sentinel-1 actually imaged each corridor, Bab-el-Mandeb runs 81% below the benchmark set by uncontested corridors, the Gulf of Aden 81% below and the Southern Red Sea 66% below. Meanwhile the Cape of Good Hope corridor carries 21.5 large vessels per imaged scene against 19.4 for the entire Suez/Red Sea route.

Two measurement traps had to be avoided to make that statement. Throughput cannot be counted using the AIS vessel category, because dark vessels have no category - in a corridor where most traffic is dark the commercial count collapses by construction and concealment is misread as evacuation. And scene counts are inferred from scenes that produced at least one detection, so a genuinely empty corridor under-counts its own observation opportunities and its per-scene rate is flattered. Both corrections push against the finding, which makes the measured deficit a lower bound.

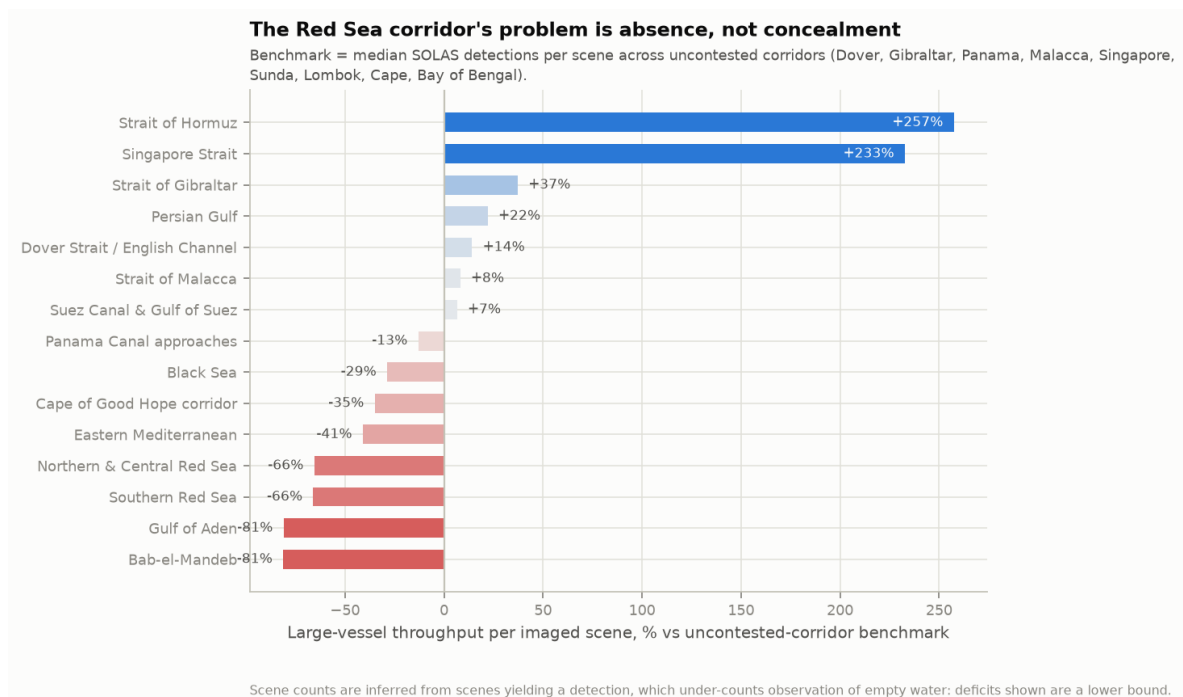


Figure 7. Large-vessel throughput per imaged scene against the uncontested-corridor benchmark. Hormuz sits far above the benchmark - its ships are present and concealed; the Red Sea corridors sit far below - their ships are elsewhere.

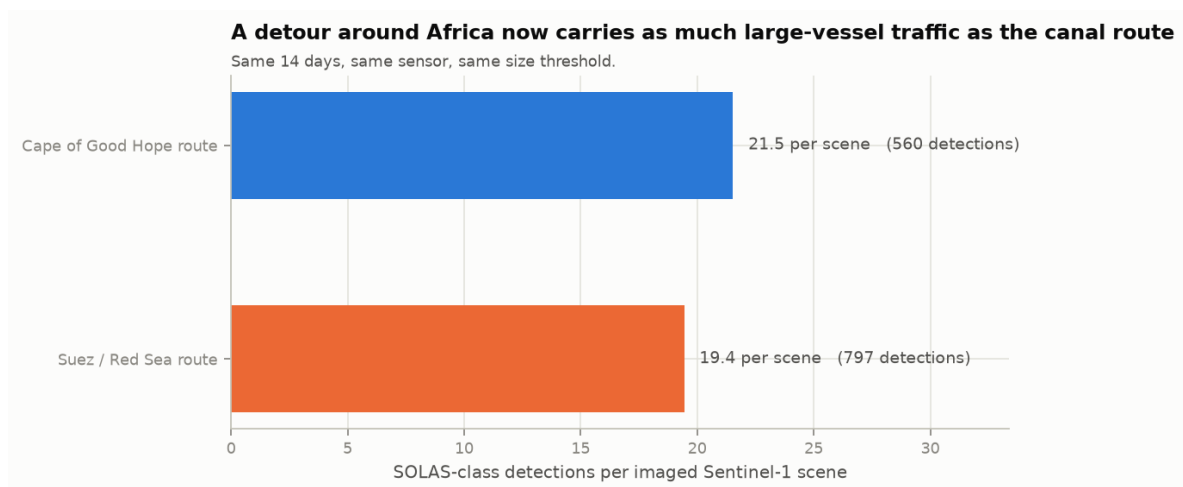


Figure 8. The Suez/Red Sea route against the Cape of Good Hope route, same sensor and same fortnight.

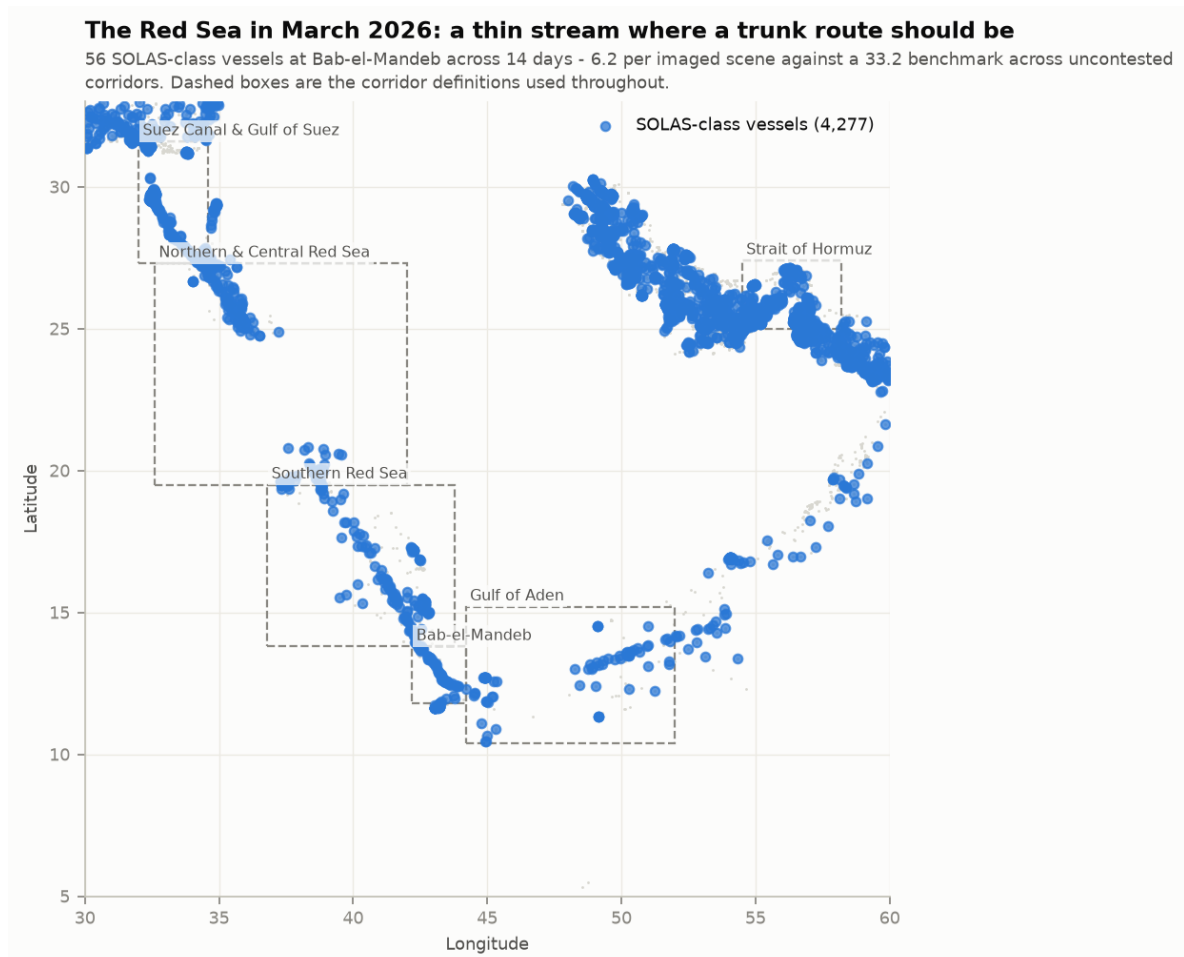


Figure 19. The Red Sea corridor with the analysis boxes drawn in. Traffic thins markedly south of Jeddah and the Bab-el-Mandeb narrows carry a trickle by the standards of a trunk route.

5.7 Is this conflict, or just geography?

Co-location is not causation, and both Hormuz and the Black Sea have sanctions regimes, dark-fleet activity and GNSS interference that are related to conflict without being conflict. Three independent tests were run rather than one, and they do not all agree - which is itself the honest result.

- **At the level of the individual ship** (39,876 SOLAS-class detections, standard errors clustered on 0.5-degree cells): being within 150 km of conflict in the preceding 30 days multiplies the odds of being dark by 2.39 (95% CI 1.24-4.60, $p=0.009$), and each log-unit of nearby conflict events adds 1.12x. Hull length pulls the other way (0.65x per log-metre), exactly as the carriage rules predict.
- **At corridor level** the association is positive but not significant (Spearman $\rho=0.28$, $p=0.18$). With 25 corridors this test has little power, and reporting it as though it confirmed the first would be dishonest. It is reported because a reader is entitled to know that the coarse-grained version of the test does not clear the bar.
- **Over time**, littoral violence and attacks at sea are contemporaneous: the cross-correlation peaks at lag 0 weeks ($r=0.63$) and is nearly flat from -10 to +10. There is no lead to exploit. Maritime posture cannot be sequenced behind a land-violence indicator.

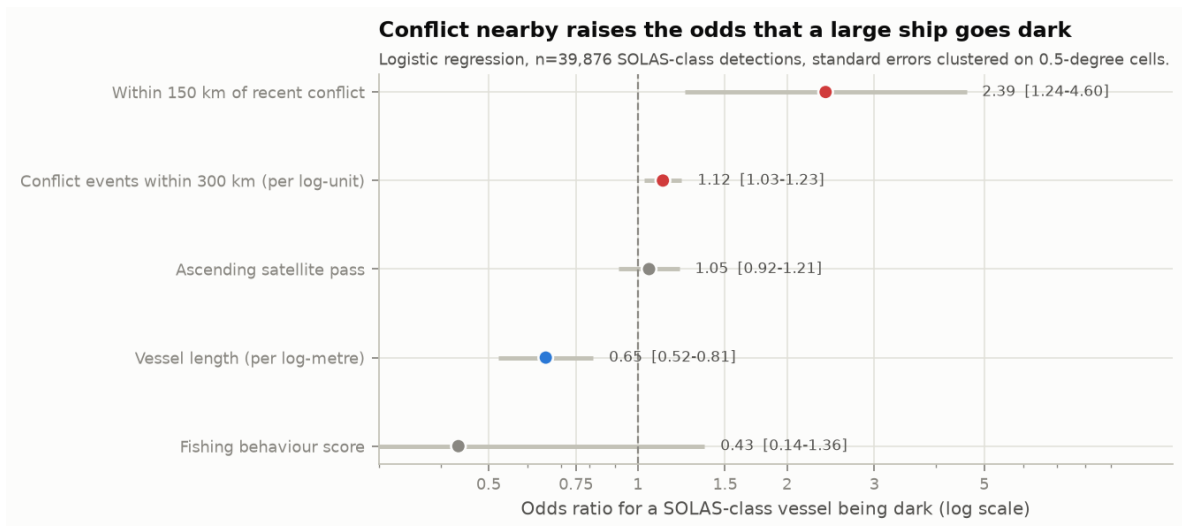


Figure 9. Odds ratios with 95% confidence intervals. Effect size is shown, not only significance: with 40,000 observations significance is cheap.

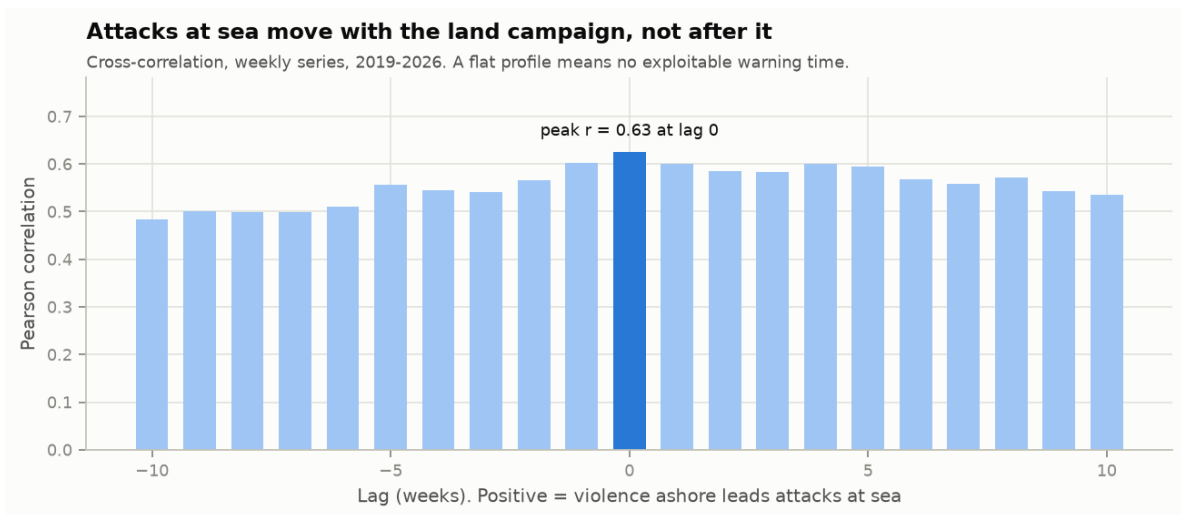


Figure 10. Cross-correlation of littoral land violence against maritime-domain events, weekly, 2019-2026.

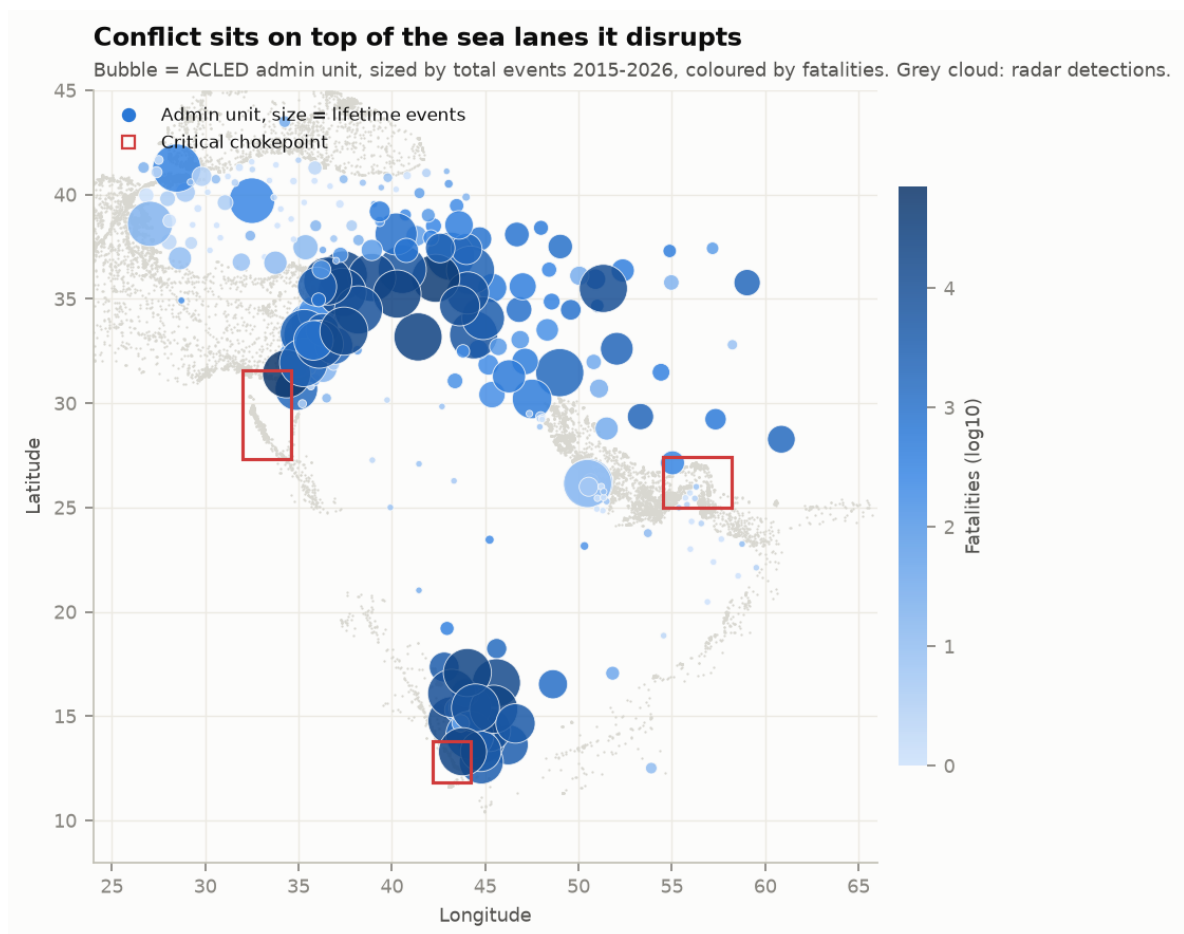


Figure 20. Conflict intensity ashore against the radar-detected sea lanes it overlooks. Red boxes mark critical chokepoints.

5.8 Predicting where AIS will go dark

Observation is not anticipation. A gradient-boosted classifier was trained to identify behavioural dark spots from cell-level features - traffic composition, density, latitude, distance to the nearest critical chokepoint, and recent conflict exposure - and validated on whole 5-degree spatial blocks held out of training. It reaches ROC-AUC 0.80 and PR-AUC 0.35 against a base rate of 0.09, i.e. roughly a fourfold lift over guessing on water the model has never seen. Distance to the nearest recent conflict is the second most important feature by permutation importance, after absolute latitude - which is itself a warning that satellite AIS reception geometry is part of what the model has learned, and one of the reasons the typology in 5.4 matters.

Dark-spot classifier: spatially blocked validation

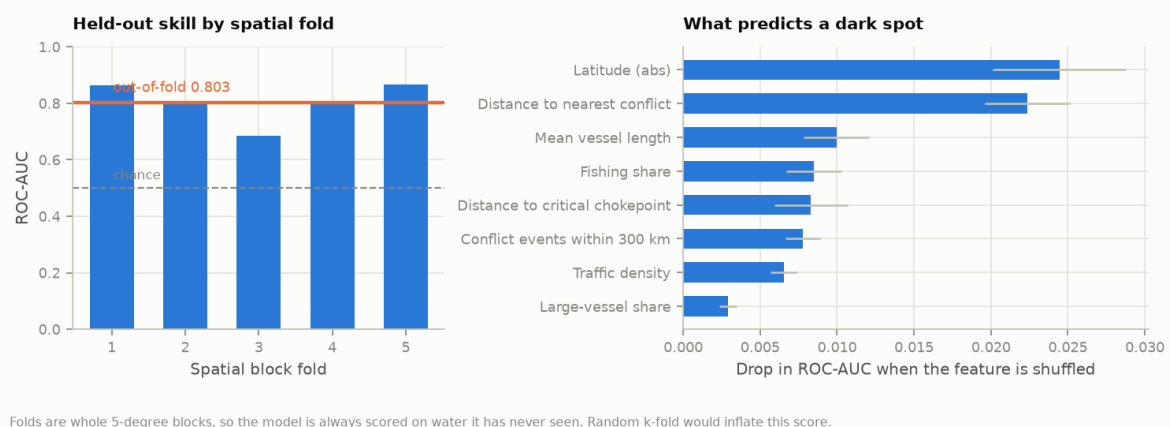


Figure 12. Spatially blocked validation and permutation importance. A random k-fold split on gridded data would have reported a much higher and much less meaningful score.

Conflict pressure on each corridor was separately forecast twelve weeks ahead. Four models competed on expanding-window backtests: the winner differs by corridor, and for the Strait of Hormuz no model beat the naive benchmark - conflict there arrives in spikes that a weekly model cannot anticipate. Reporting that, rather than quietly presenting the best-looking fit, is what makes the corridors where a model *did* win (Bab-el-Mandeb, 20% better than naive; the Black Sea and Suez, around 3-10%) worth acting on.

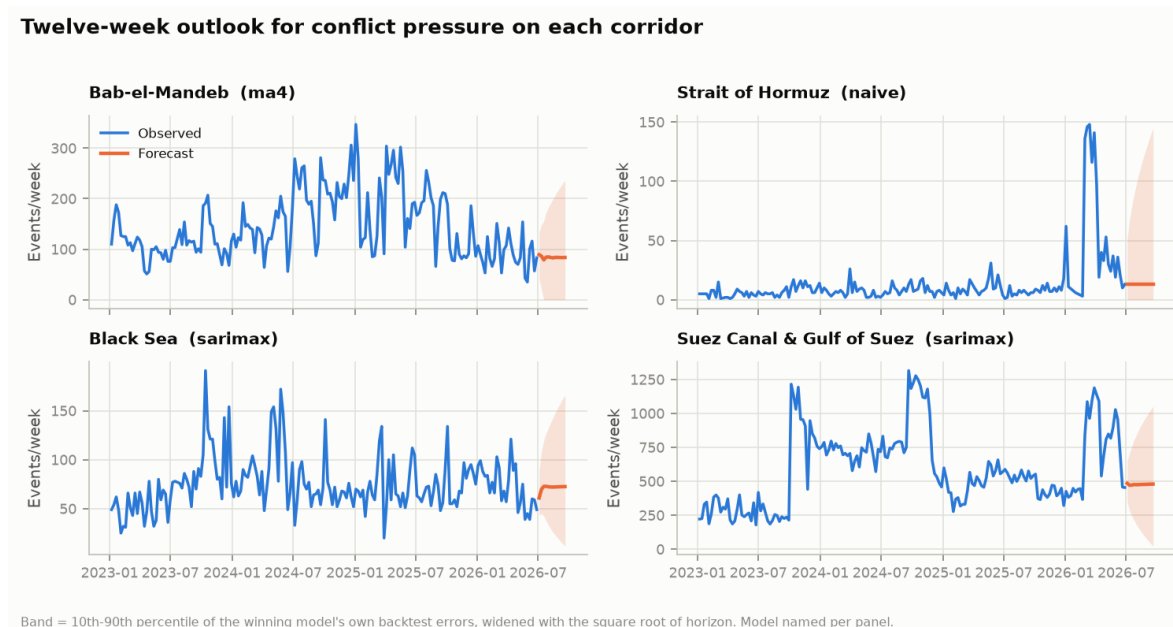


Figure 11. Twelve-week outlook for weekly conflict pressure per corridor, with intervals from each winning model's own backtest errors.

5.9 One ranked picture: the Maritime Supply-Chain Risk Index

Individual findings do not tell a staff which lane to address first. The MSRI combines conflict intensity, escalation trend, identification risk, throughput disruption and strategic criticality into a single relative ranking, with the weights declared in a config file so that an assessor who disagrees can change them and re-run rather than argue with a black box. The index is explicitly relative: 0.72 means 'among the worst lanes observed this period', never 'a 72% chance of closure'.

Rank	Corridor	MSRI	Tier	Driver	Dark %	Throughput	India exp.
1	Gulf of Aden	0.72	RED	Throughput	19%	-81%	0.54
2	Southern Red Sea	0.70	RED	Conflict	23%	-66%	0.56
3	Bab-el-Mandeb	0.68	RED	Throughput	5%	-81%	0.54
4	Black Sea	0.67	RED	Conflict	48%	-29%	0.30
5	Eastern Mediterranean	0.66	RED	Conflict	8%	-41%	0.26
6	Strait of Hormuz	0.62	RED	Identification	84%	+257%	0.62
7	Gulf of Oman	0.59	AMBER	Conflict	46%	-27%	0.53
8	Suez Canal & Gulf of Suez	0.54	AMBER	Conflict	3%	+7%	0.44

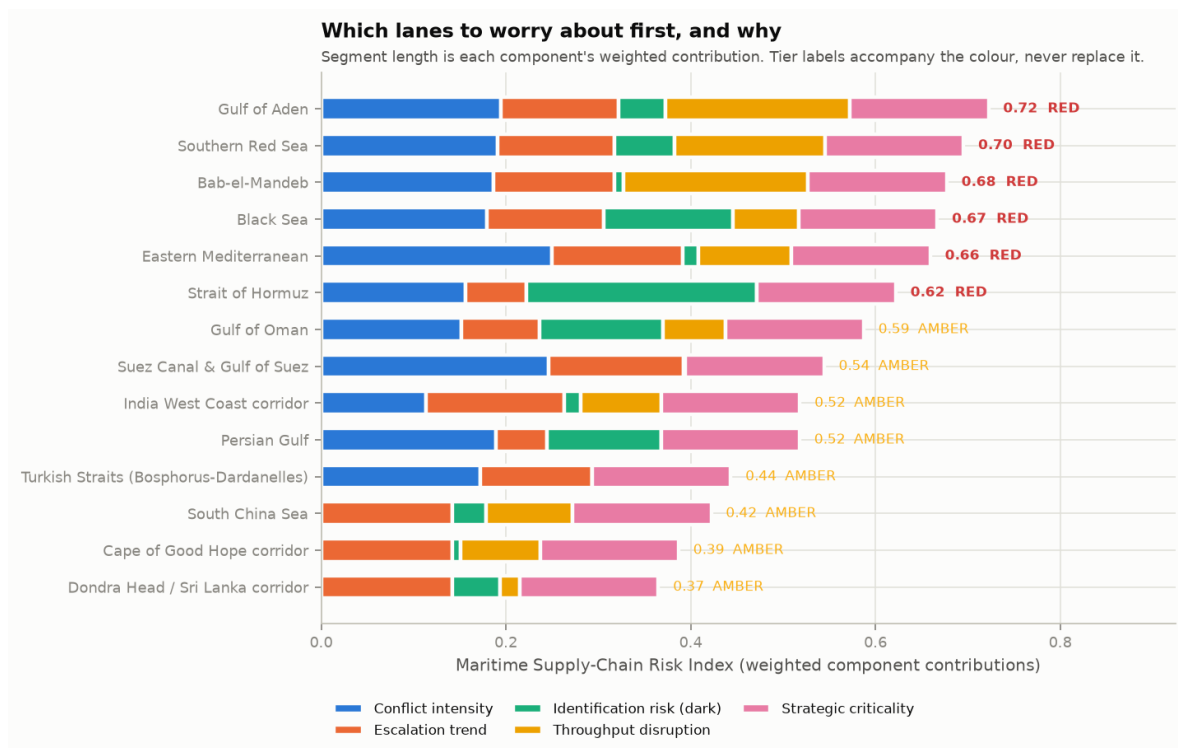


Figure 13. MSRI with each component's weighted contribution shown, so the ranking can be argued with rather than accepted.

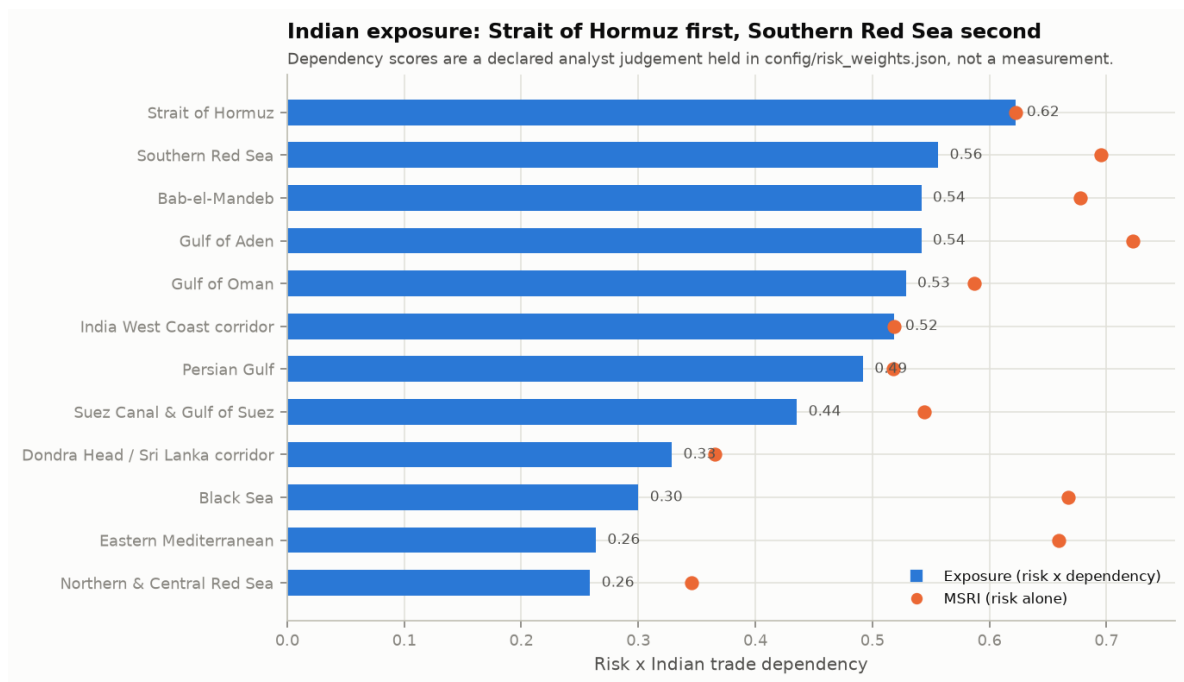


Figure 14. Risk weighted by Indian trade dependency. Dependency scores are a declared analyst judgement (config/risk_weights.json), not a measurement, and are listed with their rationale in Appendix B.

6. What this means, and what to do about it

6.1 For the recognised maritime picture

- **Make the dark rate an indicator, not an anecdote.** A SOLAS-class dark rate per corridor, computed weekly from one SAR product and one AIS feed, is a measurable and reportable warning indicator. The threshold suggested by this data is 35%: below it corridors sit in a tight band around 2-20%; above it they are outliers by a wide margin and every one of them is in a contested theatre.
- **Report the size-controlled number or none at all.** An uncontrolled dark rate will make the Bay of Bengal look worse than the Black Sea, and a staff that is briefed that once will discount the indicator thereafter.
- **Classify before tasking.** Dead zone, non-carriage and behavioural darkness demand different responses - a receiver or feed fix, no action, and collection respectively. The discriminator is free: does anything else in that cell match?

6.2 For Indian shipping and naval planning

- **Hormuz is an identification problem.** At 84% dark, AIS-derived traffic pictures of this strait should be treated as structurally incomplete rather than merely noisy. Practical consequences: independent (radar or space-based) tracking for Indian-flag and Indian-crewed transits, pre-sailing and in-transit reporting that does not depend on the vessel's own transponder, and an assumption that any AIS-based picture of the Gulf is under-counting large hulls by a factor of several.
- **The Red Sea is a routing problem.** With throughput 81% below peer corridors and the Cape route carrying comparable large-vessel density, the planning question is no longer whether diversion happens but what a sustained Cape routing costs in hull-days, bunker, schedule reliability and escort geography - and what that implies for stock policy on crude, fertiliser and containerised imports.
- **The Gulf of Oman is the early-warning water.** It carries the second-highest India exposure score, sits outside the strait itself, and already shows 46% dark among large hulls. It is where a deterioration at Hormuz becomes visible first while sea room still exists.
- **Watch the Sri Lanka corridor and the west-coast approaches for the same signature.** Both currently sit in the normal band. A rise in the SOLAS-class dark rate there, with no corresponding conflict, would indicate the dark-fleet practice migrating into Indian near waters - a different and slower-moving threat than any attack, and one only this measurement finds.

6.3 For the analytics function itself

- **Fuse, do not choose.** Neither dataset supports any conclusion in this report alone. AIS alone cannot see what it is not told; SAR alone cannot tell a tanker from a tanker that is lying about itself.
- **Publish the weights.** Every judgement in the MSRI is in a config file. An index whose weights are invisible cannot be challenged by the commander it advises, and so will not be believed by them.
- **Automate the fortnight, not the year.** Everything here runs in under two minutes on one core from two flat files. It is a weekly product, not a study.

7. Limitations - what would change these conclusions

Limitation	Effect on conclusions	How it was mitigated / what would settle it
The detection window is 14 days	No before/after comparison is possible	Comparison is made against uncontested corridors rather than before/after
Sentinel-1 does not image all scenes	Daily detection counts partly measure unimaged areas	Adjusted for unimaged areas; the residual bias under-counts
Scene footprints are not in the corridor	Observation opportunity is lost	Stated explicitly, the jurisdiction is established and runs against the finding
Conflict data is Middle East only	Coupling tests cannot be run for Asia	Those corridors serve as the control group, which the design requires anyway
ACLED locates events at admin level	Distances are inaccurate	Exposure radii are deliberately wide (150/300/600 km); no claim is made at finer scale
Maritime ACLED events carry only one coordinate	No positional information	Used only for timing (the lead-lag test), never for position
AIS matching is the provider's, not system's	Systematic provider bias would propagate	Dark definitions are carried through every corridor result; the ranking is sensitive to dark
Dark is not proven deliberate	Sanctions evasion, GNSS jamming, etc.	Each type of separate conduct is produced from selective behaviour, but intent is not proven
MSRI weights are judgement	A different weighting reorders the model	High tier is internal, declared, and documented; the RED tier is robust to reasonable change

8. Reproducibility and the Power BI deliverable

The full pipeline runs from the two supplied files with `python run.py --all` and writes every figure, every table behind every figure, a run manifest recording the settings and row counts, and the Power BI export. No network access is required and no GIS stack is installed: the maps are drawn from the detections themselves.

The submission format asks for a .pbix. A .pbix is a binary artefact only Power BI Desktop can author, so what the pipeline produces is what a .pbix is built from and the instructions to assemble it in minutes: a star schema (one CSV per table, dimensions and facts separated, a single date dimension driving both the weekly conflict facts and the daily detection facts), a correctly modelled many-to-many bridge between admin units and corridors so that events are not double-counted, `measures.dax` with the DAX for every measure used, `relationships.csv` specifying every join, and `docs/POWERBI_GUIDE.md` with the page-by-page build.

Deliverable	Location	Contents
This report	outputs/report/	PDF submission
Figures	outputs/figures/	20 PNG figures at 160 dpi
Audit tables	outputs/tables/	One CSV behind every figure and every claim
Power BI model	outputs/powerbi/	14 star-schema CSVs, <code>measures.dax</code> , <code>relationships.csv</code>
Run manifest	outputs/run_manifest.json	Settings, row counts, runtime, headline values for the run that produced this document
Source code	src/dcsc/	Documented pipeline, ~2,500 lines, with tests

Appendix A. Dark operating areas recovered by clustering

DBSCAN (60 km neighbourhood, 40-contact minimum, haversine metric) over SOLAS-class dark detections. 'Local dark rate' is computed over *all* large-vessel detections within the cluster's own extent, so a high value means matched traffic was present and still the cluster stayed dark.

Cluster	Corridor	Dark contacts	Centre	Mean length	Local dark	Extent
5	Strait of Hormuz	1,401	25.6N 55.6E	184 m	82%	176 km
2	Black Sea	510	44.7N 37.3E	163 m	76%	136 km
6	Persian Gulf	134	25.8N 51.9E	186 m	62%	52 km
8	Open ocean / other	120	36.1N 14.7E	174 m	69%	95 km
9	Open ocean / other	98	35.4N 139.8E	138 m	100%	69 km
1	Black Sea	63	46.9N 38.3E	128 m	70%	55 km
7	Persian Gulf	58	27.5N 52.1E	174 m	81%	55 km
10	Bay of Bengal	55	21.0N 87.7E	186 m	37%	67 km
3	Open ocean / other	55	54.7N 18.5E	155 m	54%	71 km
0	Open ocean / other	48	36.3N 22.7E	167 m	59%	61 km

Appendix B. Declared India-dependency judgements

These scores are an analyst judgement, not a measurement, and they affect only the India overlay - never the MSRI itself. They are held in config/risk_weights.json so that they can be edited and the analysis re-run.

Corridor	Dependency Rationale	
Strait of Hormuz	1.00	Bulk of India's crude and LPG imports originate inside the Gulf and must transit Hormuz.
Southern Red Sea	0.80	Same corridor, northbound leg.
Bab-el-Mandeb	0.80	Suez routing for India-Europe and India-US East Coast trade.
Gulf of Aden	0.75	Convoy and escort water for Indian-flag shipping; anti-piracy patrol area.
Gulf of Oman	0.90	Immediate approach to Hormuz; Indian-flag and Indian-crewed tankers transit daily.
India West Coast corridor	1.00	Kandla, Mumbai, JNPT, Cochin approaches - India's own gateway waters.
Persian Gulf	0.95	Loading ports for the same crude and LPG flows.
Suez Canal & Gulf of Suez	0.80	India-Europe container and refined-product flows.
Dondra Head / Sri Lanka corridor	0.90	Main east-west lane south of Sri Lanka; India's container transshipment axis.
Black Sea	0.45	Sunflower oil and fertiliser imports; a supply concentration rather than a route India sails.
Eastern Mediterranean	0.40	Onward leg of the Suez route to European markets.
Northern & Central Red Sea	0.75	Same corridor, approach to Suez.

Appendix C. Settings used for this run

Parameter	Value	Meaning
presence_score_floor	0.9	Detection confidence gate
min_length_m	10.0	Minimum reliable detection size
match_score_floor	1.0	Weak-match threshold for the strict dark definition
grid_deg	0.5	Dark-rate surface cell size (degrees)
min_cell_detections	25	Minimum traffic before a cell is tested
dbscan_eps_km	60.0	Cluster neighbourhood radius
dbscan_min_samples	40	Minimum contacts forming a cluster

Parameter	Value	Meaning
coastal_buffer_km	250.0	Conflict-to-corridor attribution radius
forecast_horizon_weeks	12	Forward forecast length
backtest_weeks	26	Expanding-window backtest length
random_state	20260930	Seed for every stochastic step

Generated 2026-09-12 16:28 UTC by the pipeline in this repository. Every figure is reproducible with `python run.py --all`; every number in this document is read from the frames that produced the figures beside it.